Online Retail Customer Churn Analysis Report

1. Problem Understanding

Customer churn is a critical metric for online retailers. It represents the percentage of customers who stop doing business with the company. Identifying potential churners allows the business to implement targeted retention strategies, thereby reducing customer acquisition costs and increasing long-term profitability. This report analyzes historical customer data to understand the drivers of churn and provides segmentation-based insights for targeted marketing.

2. Approach

The analysis was conducted in four main phases:

- Data Preprocessing: Handling categorical variables and scaling numerical data.
- Exploratory Data Analysis (EDA): Visualizing distributions and correlations.
- Customer Segmentation: Using K-Means clustering to group customers based on spending and income.
- Churn Prediction: Training and comparing classification algorithms (Logistic Regression, Random Forest, Gradient Boosting).

3. Graphs and Visualizations

The following visualizations provide insights into customer behavior and churn patterns.

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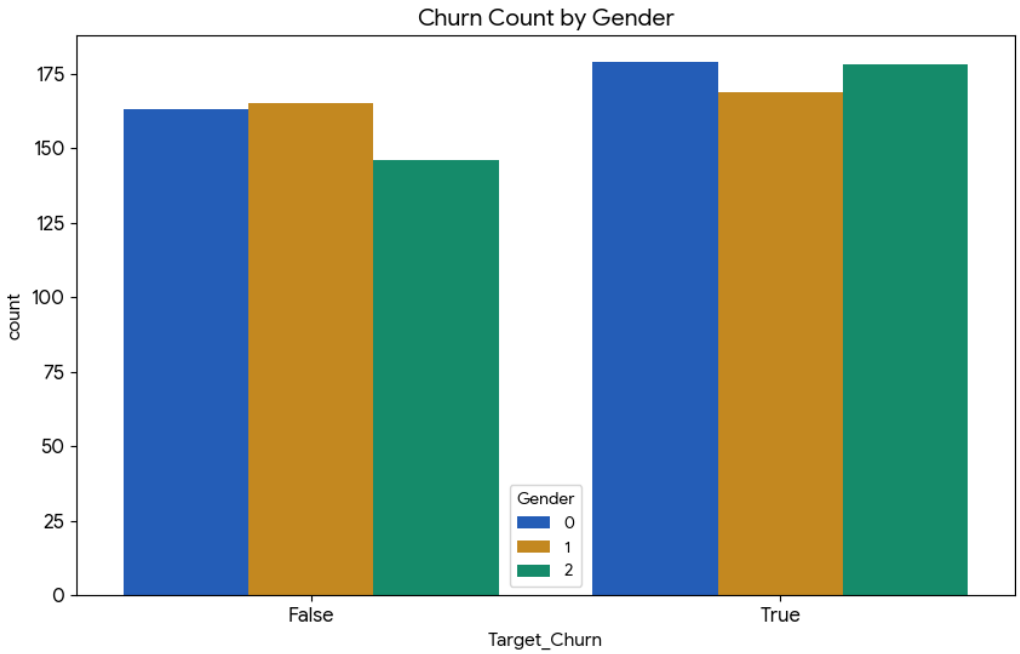

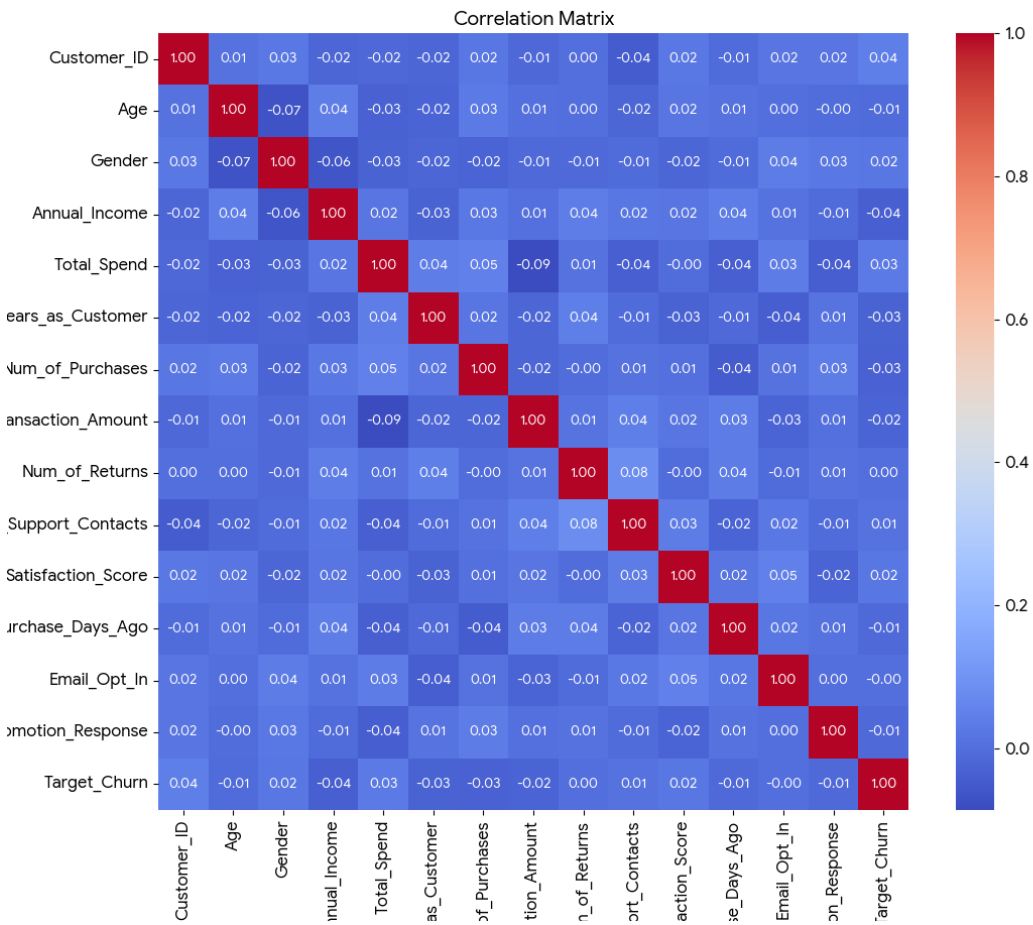

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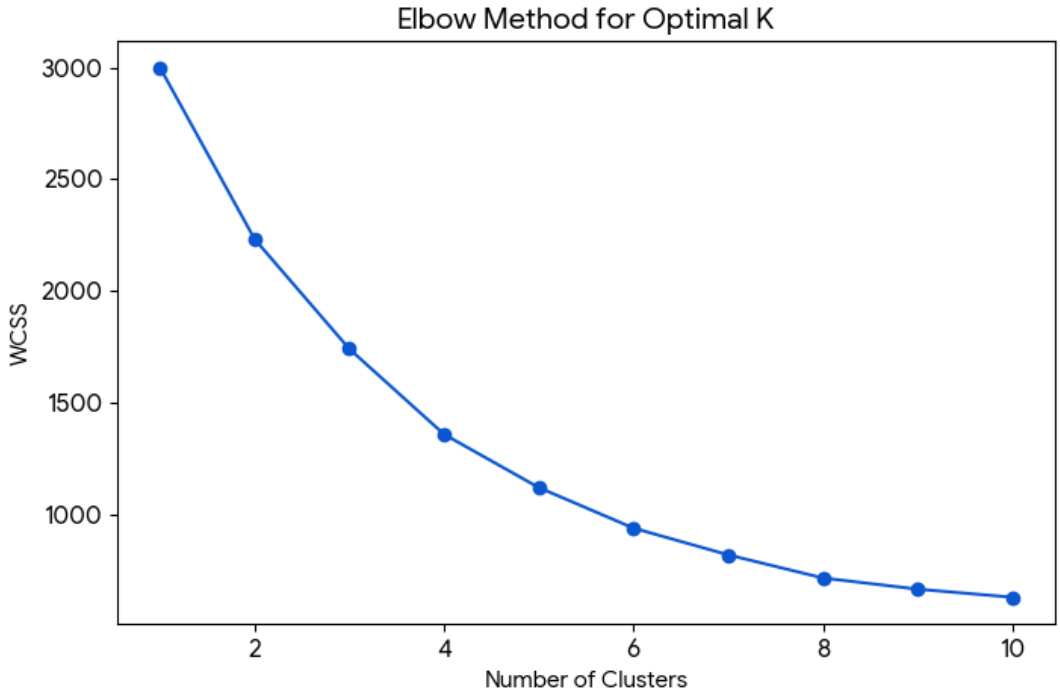
4. Customer Segmentation (Clustering)

Customers were segmented using K-Means clustering based on Annual Income, Total Spend, and Number of Purchases. The elbow method suggests an optimal cluster count. We identified 4 distinct customer segments based on their economic behavior.



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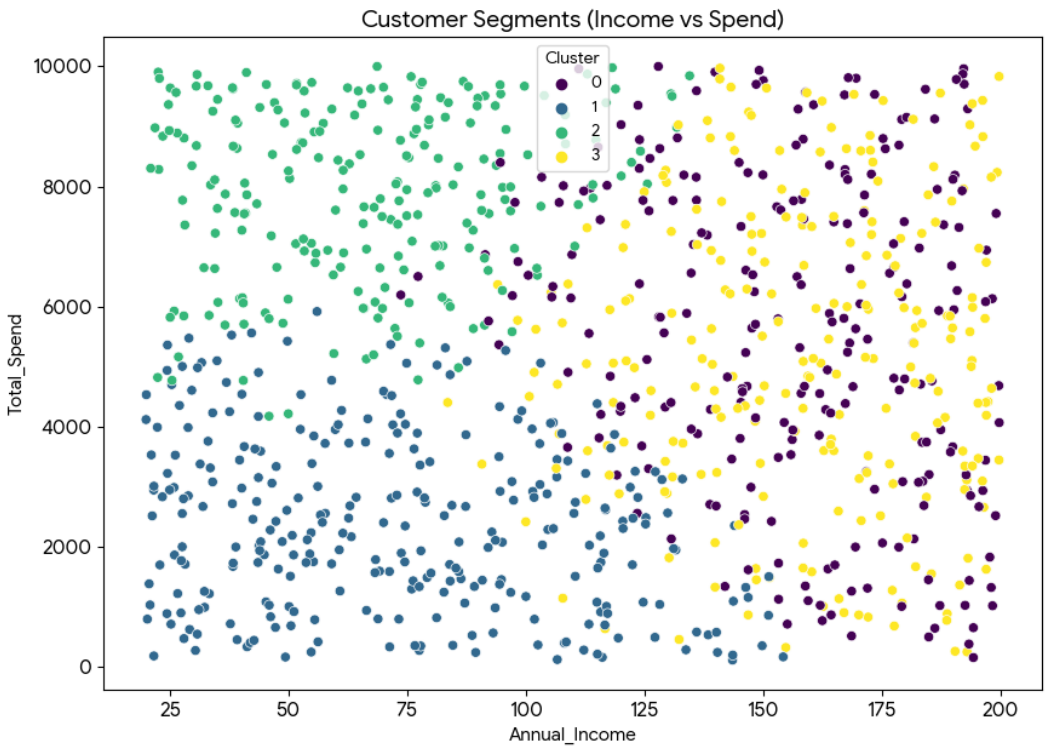

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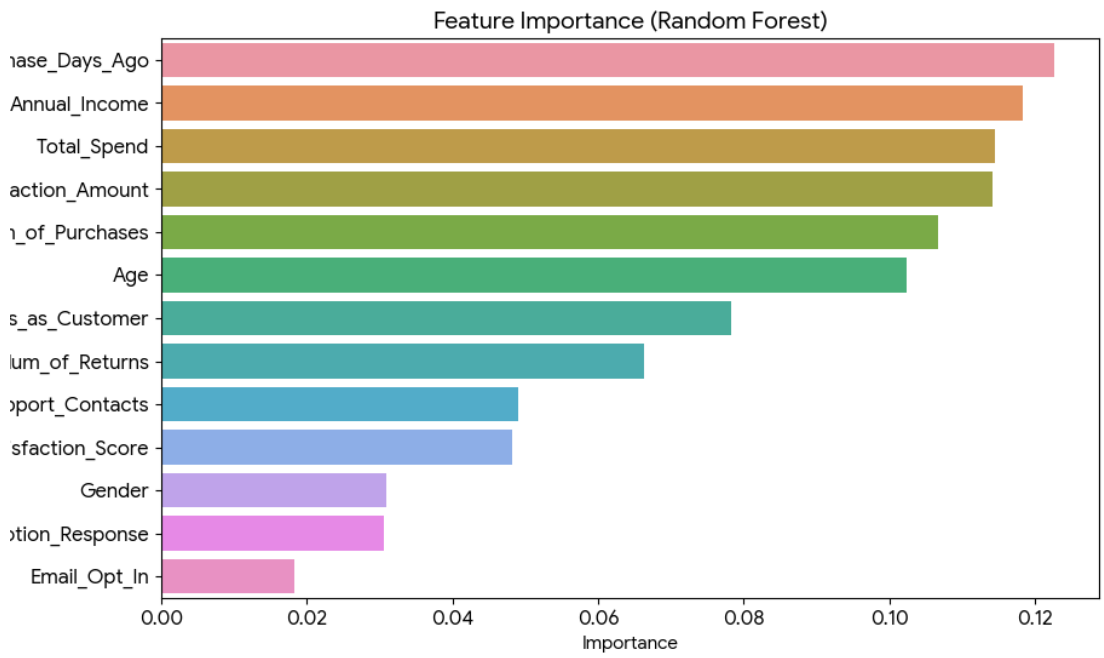
5. Algorithm Comparison

Comparison of performance metrics across different classification models:

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.455	0.490	0.689	0.573
Random Forest	0.560	0.569	0.698	0.627
Gradient Boosting	0.560	0.570	0.689	0.624

Feature importance analysis from the Random Forest model highlights which factors most influence churn.



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6. Business Recommendations

Based on the analysis, the following actions are recommended:

1. High-Value Retention: Focus on 'High Income' customers. Those with high spend but low frequency should be targeted with loyalty rewards to increase their visit frequency.

2. Improve Customer Satisfaction: Satisfaction scores and support contacts are significant predictors. Implementing a rapid response team for customers with low satisfaction scores could prevent immediate churn.

3. Re-Engagement Campaigns: Target customers with high 'Last Purchase Days Ago' values with personalized discounts to bring them back.

4. Model Refinement: The current models show moderate predictive power. Collecting more behavioral data could improve prediction accuracy.